Spatial Digital Twin for Unsupervised Wildfire Detection with GOES-16 ABI: Multi-Band Pipeline, Temporal Assimilation, Multi-View Consensus, and Agentic Orchestration

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https://github.com/naubergois/ceara-queimadas

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v3.1 — Updated literature survey (GraphFire-X, Thermodynamic GeoAI, FireCastNet), FIRMS validation, expanded discussion

Abstract

Wildfire detection in Brazilian biomes—Amazon, Cerrado, Pantanal, and Caatinga—remains a critical challenge, lacking operational digital twins that integrate real-time satellite data, physical simulation, and autonomous decision support. This paper presents an unsupervised spatial digital twin framework for wildfire detection using GOES-16 ABI L2 CMIPF data over the state of Ceará, Brazil. The system combines: (i) multi-scale median filtering of thermal band T_B13 (10.3 μm) with weighted spectral contrast T_B7−T_B14 (3.9−11.2 μm); (ii) temporal assimilation via probabilistic fusion with exponential decay (`prob_or`); (iii) a combined persistence method fusing peak intensity, temporal mean, and activation dwell ratio across hourly granules; (iv) a consensus ensemble requiring agreement of 2 of 3 physical views (digital twin, persistence, spatial residual); and (v) a LangGraph-based agentic orchestration pipeline with ReAct reasoning, spatial cross-referencing, and auditable alert generation. Evaluation against 76 INPE reference fire foci on 2024-10-31 with automatic contamination calibration maximizing F1 yields a consistent performance hierarchy: digital twin (P=0.032, R=0.106, F1=0.049), isolation forest (P=0.026, R=0.063, F1=0.037), combined persistence (P=0.031, R=0.007, F1=0.012), spatial residual (P=0.019, R=0.004, F1=0.006), and consensus ensemble (P=0.000, F1=0.000). Synthetic benchmarks confirm all methods exceed precision > 0.8 and F1 > 0.8 under well-posed grid-aligned conditions (8/8 test assertions pass). We identify four structural misalignments—temporal, semantic, scale, and product—as root causes of the synthetic-real gap. Extending this diagnostic foundation, we introduce the Neural Koopman Physics-Informed Graph Neural Network (NeKo-PIGNN) mathematical framework for predictive fire propagation modeling, combining Koopman operator theory (absolute gap in wildfire literature) with physics-constrained GNNs regularized by the Rothermel equation. A LangGraph pipeline orchestrates 9 specialized AI agents for data collection, geospatial analysis, climate risk assessment, GOES-16 persistence analysis, evidence fusion, ReAct diagnosis, alert generation, technical reporting, and auditing—all integrated with a React/TypeScript frontend and PostGIS spatial database. We propose a complete six-layer roadmap for a bidirectional Brazilian fire digital twin integrating INPE, NASA FIRMS, MapBiomas, IoT sensor networks, and autonomous agentic decision support.

Resumo

A detecção de incêndios florestais nos biomas brasileiros—Amazônia, Cerrado, Pantanal e Caatinga—continua sendo um desafio crítico, sem gêmeos digitais operacionais que integrem dados de satélite em tempo real, simulação física e suporte autônomo à decisão. Este artigo apresenta um framework não supervisionado de gêmeo digital espacial para detecção de queimadas usando dados GOES-16 ABI L2 CMIPF sobre o estado do Ceará, Brasil. O sistema combina: (i) filtragem mediana multi-escala da banda térmica T_B13 (10,3 μm) com contraste espectral ponderado T_B7−T_B14 (3,9−11,2 μm); (ii) assimilação temporal via fusão probabilística com decaimento exponencial (prob_or); (iii) um método de persistência combinada fundindo intensidade de pico, média temporal e razão de ativação entre granules horários; (iv) um ensemble de consenso exigindo concordância de 2 em 3 visões físicas (gêmeo digital, persistência, residual espacial); e (v) um pipeline de orquestração agentiva baseado em LangGraph com raciocínio ReAct, cruzamento geoespacial e geração auditável de alertas. A avaliação contra 76 focos de referência do INPE em 31/10/2024 com calibração automática de contaminação maximizando F1 produz uma hierarquia de desempenho consistente: gêmeo digital (P=0,032, R=0,106, F1=0,049), isolation forest (P=0,026, R=0,063, F1=0,037), persistência combinada (P=0,031, R=0,007, F1=0,012), residual espacial (P=0,019, R=0,004, F1=0,006) e ensemble de consenso (P=0,000, F1=0,000). Benchmarks sintéticos confirmam que todos os métodos excedem precisão > 0,8 e F1 > 0,8 sob condições alinhadas (8/8 testes aprovados). Identificamos quatro desalinhamentos estruturais—temporal, semântico, de escala e de produto—como causas raiz da lacuna sintético-real. Estendendo esta base diagnóstica, introduzimos o framework matemático Neural Koopman Physics-Informed Graph Neural Network (NeKo-PIGNN) para modelagem preditiva de propagação de fogo, combinando a teoria do operador de Koopman (lacuna absoluta na literatura de queimadas) com GNNs com restrição física regularizadas pela equação de Rothermel. Um pipeline LangGraph orquestra 9 agentes especializados de IA para coleta, análise geoespacial, risco climático, persistência GOES-16, fusão de evidências, diagnóstico ReAct, geração de alertas, relatoria técnica e auditoria—tudo integrado com frontend React/TypeScript e banco de dados espacial PostGIS. Propomos um roteiro completo em seis camadas para um gêmeo digital bidirecional brasileiro de queimadas integrando INPE, NASA FIRMS, MapBiomas, redes de sensores IoT e suporte autônomo à decisão agentiva.

Keywords: Digital Twin; Wildfire Detection; GOES-16; Unsupervised Learning; Remote Sensing; Brazilian Biomes; Multi-View Consensus; Koopman Operator; Physics-Informed GNN; LangGraph; Agentic AI

Palavras-chave: Gêmeo Digital; Detecção de Queimadas; GOES-16; Aprendizado Não Supervisionado; Sensoriamento Remoto; Biomas Brasileiros; Consenso Multi-Vista; Operador de Koopman; GNN com Informação Física; LangGraph; IA Agêntica

1 Introduction

1.1 Context and Motivation

Wildfires in Brazil pose a growing threat to globally significant ecosystems. The Amazon—Earth's largest terrestrial carbon sink—recorded over 110,000 fire foci in 2024 alone [4]. The Cerrado, the most biodiverse savanna on the planet, concentrates the highest number of active fire detections nationally, with recurrent fires destroying vast tracts of native vegetation annually. The Pantanal suffered catastrophic fires in 2020 and 2024, destroying over 30% of the biome [11]. In the Caatinga, an exclusively Brazilian semi-arid biome covering approximately 70% of the state of Ceará, anthropogenic fires for land management and agricultural expansion are increasingly severe during prolonged drought periods.

Despite this urgency, Brazil does not currently operate a digital twin for wildfire detection and response [14, 18]. Existing monitoring systems—INPE Queimadas, PRODES, DETER, BDQueimadas, and MapBiomas Fogo—provide essential fire alert and deforestation monitoring services. However, these are fundamentally read-only monitoring platforms, not bidirectional digital twins as defined by ISO/IEC 23247 [6]. A systematic survey of digital twin technologies for fire detection in Brazilian biomes [15, 17] confirms that no operational digital twin exists for Brazilian wildfire detection—all current programs are alert/monitoring systems lacking predictive simulation, what-if scenario modeling, bidirectional control loops, and AI-driven autonomous decision support.

The state of Ceará, in northeastern Brazil, presents a compelling case study. Its territory spans 148,920 km^2 with diverse eco-regions from the humid coastal strip (Mata Atlântica remnants) to the semi-arid Sertão (Caatinga biome) and the chapadas of the Ibiapaba and Araripe plateaus. Fire activity peaks between August and December, coinciding with the dry season when relative humidity drops below 30% and cumulative rainfall deficits exceed 200 mm. The availability of near-real-time GOES-16 data, operational INPE fire foci records, and a well-defined administrative geography makes Ceará an ideal proving ground for a spatial digital twin prototype.

1.2 Related Work

International research has advanced digital twin-based wildfire management through several distinct architectures:

- **IVSR** [14]: A bidirectional digital twin with autonomous AI agents, integrating multi-sensor imagery, meteorological data, and 3D forest models with similarity-based scenario alignment. Demonstrates significant reductions in detection-to-intervention latency through closed-loop UAV and team reallocation.
- **FIRE-VLM** [24]: The first vision-language-model-guided reinforcement learning framework in a physics-grounded fire digital twin, achieving up to 6× detection speedup using UAVs with dual-view sensing. The GIS-to-simulation pipeline enables automatic construction of fire digital twins from remote sensing data.
- **FIRETWIN** [18]: A cyber-physical digital twin using Unreal Engine + CAWFE model for King Fire reconstruction, providing immersive interactive analysis and multi-modal sensing integration.
- **AIMNET** [26]: An IoT-empowered digital twin for continuous gas emission monitoring with coupled weather-transport modeling, using a three-layer architecture (physical world → bidirectional links → digital world).
- **H-RDT** [21]: A hazard-responsive digital twin using Physics-Informed Neural Networks for the fire-power-outage-urban-heat-wave cascade.
- **PINNs for fire spread** [23]: Physics-informed neural networks learning wildfire propagation parameters, providing a foundation for data-driven digital twins with physical consistency.
- **Neural ODEs** [2]: Continuous-time dynamics modeling that naturally accommodates irregularly timed satellite observations.
- **Koopman operator theory** [1, 7]: A framework for linearizing nonlinear dynamics through observable functions, enabling spectral analysis and optimal control.
- **GraphFire-X** [3]: A physics-informed graph attention network ensemble for building-scale wildfire vulnerability at the wildland-urban interface (WUI). Combines dual-expert GNN (fire physics-informed convection, radiation, ember probability attention) with XGBoost for structural fragility assessment. Validated on the 2025 Eaton Fire. Uses Google AlphaEarth embeddings and community-level graph-of-contagion representation. The physics-informed GNN approach is directly comparable to our NeKo-PIGNN framework, though GraphFire-X targets building vulnerability rather than satellite-based fire propagation.
- **Thermodynamic GeoAI** [10]: A thermodynamic-inspired explainable GeoAI framework using statistical physics equilibrium (Burden/Capacity) for detecting wildfire-induced PM2.5 anomalies. Validated on the 2023 Canadian wildfires. Provides a complementary approach for smoke plume analysis in Brazilian biomes, identifying phase transitions in spatial systems and causal regime shifts.
- **FireCastNet** [13]: An Earth-as-a-Graph framework combining 3D Conv with GraphCast GNN for global seasonal fire prediction. Validated on the SeasFire dataset—a multivariate Earth System data cube. Outperforms GRU, Conv-LSTM, U-TAE, and TeleViT baselines, with strong results over

South America. It is the first GraphCast-based fire model validated globally and provides the only operational-scale seasonal fire prediction framework applicable to Brazilian biomes.

None of these systems have been adapted to Brazilian biomes or integrated with operational INPE data flows. The specific challenges of Brazilian biomes—Amazon cloud cover, continental scale (8.5M km^2), limited connectivity in remote areas, and the need to distinguish controlled burns from criminal fires—remain unaddressed by existing international digital twin frameworks. Recent work specific to Brazilian biomes includes deep learning fire detection on Landsat/Sentinel-2 imagery over the Amazon [22], network science analysis of Brazilian wildfire patterns [12], and high-resolution digital twin wildfire simulation using 1 m DEM data [9], yet none proposes a complete digital twin architecture with bidirectional control.

1.3 Contributions

The contributions of this work span four domains—detection methodology, mathematical modeling, agentic orchestration, and operational deployment:

1. **Unsupervised multi-scale thermal anomaly detection** combining median filtering residuals at three scales (5×5, 9×9, 15×15 pixels) in T_B13 with weighted spectral contrast (T_B7−T_B14), enabling fire detection without labeled training data.

2. **Temporal assimilation framework** (`prob_or`) fusing hourly risk scores with exponential decay, modeling thermal anomaly persistence between GOES-16 scans.

3. **Combined persistence method** fusing peak intensity, temporal mean, and activation dwell ratio with adaptive per-hour percentiles and conditional morphological filtering.

4. **Consensus ensemble** requiring 2-of-3 agreement among digital twin, persistence, and spatial residual physical views.

5. **NeKo-PIGNN mathematical framework** introducing Neural Koopman Physics-Informed Graph Neural Networks for predictive fire propagation—a combination of Koopman operator theory (absolute literature gap for wildfire) with physics-constrained GNNs regularized by the Rothermel equation.

6. **LangGraph agentic orchestration pipeline** with 9 specialized AI agents for data collection, validation, geospatial analysis, climate risk, GOES-16 persistence, evidence fusion, ReAct diagnosis, alert generation, and auditing.

7. **Systematic diagnosis** of four structural misalignments between GOES-16 CMIPF grids and INPE point foci.

8. **Six-layer roadmap** for a bidirectional Brazilian fire digital twin integrating satellite data, physical simulation, IoT sensors, 3D visualization, and autonomous agentic decision support.

2 Methodology

2.1 Data Sources and Preprocessing

We use the GOES-16 ABI L2 CMIPF (Cloud and Moisture Imagery Product) in netCDF format from the NOAA AWS Open Data bucket [16]. The CMIPF product provides calibrated brightness temperatures at 2 km resolution at nadir. For the Ceará region (bbox: min_lat = −7.9°, max_lat = −2.5°, min_lon = −41.5°, max_lon = −37.0°), this yields approximately 72×72 grid cells per channel after remapping.

Three spectral channels are employed:

Table 1: GOES-16 ABI spectral channels used.

Band	Wavelength	Purpose
Band 7	3.9 μm (SWIR)	Sensitive to sub-pixel hot spots
Band 13	10.3 μm (clean IR)	Primary temperature channel
Band 14	11.2 μm (longwave)	Atmospheric window channel

The DQF (Data Quality Flag) field retains only pixels with DQF=0 (nominal quality). For UTC hours $h \in \{16, 17, 18\}$ on evaluation date 2024-10-31, we download the nearest available scan granule per channel.

Reference fire foci are obtained from the INPE BDQueimadas API [5] for Ceará (UF code 23). The INPE set contains 76 fire foci on 2024-10-31, each with latitude, longitude, and UTC timestamp. After binning to the 72×72 GOES grid, these map to 49 raw truth cells and 284 truth cells after one iteration of 3×3 morphological dilation. This dilation is a necessary compensation for the coarse GOES grid resolution ($\sim$0.075° per cell, $\approx$56 km^2 per cell) relative to the nearly point-like INPE foci.

Additionally, the system ingests real-time data from NASA FIRMS (VIIRS and MODIS), Open-Meteo for meteorological variables, and Nominatim for reverse geocoding of fire foci to municipal boundaries. On 2026-06-06, for instance, FIRMS recorded 28 active hotspots across Ceará (VIIRS SNPP=4, NOAA20=8, NOAA21=11, MODIS=5) with a mean FRP of 4.8 MW, while the preceding day (2026-06-05) recorded 142 hotspots (VIIRS SNPP=39, NOAA20=53, NOAA21=44, MODIS=6) with mean FRP=11.2 MW and 24 severe events (FRP $\geq$ 20 MW). This day-to-day variability—a 426% increase driven largely by the Oeiras/Picos cluster (68 foci, 1223.7 MW total FRP)—highlights the dynamic nature of fire activity in the region and validates the need for continuous autonomous monitoring. The system's capacity for cross-sensor validation across four independent satellite platforms is demonstrated.

2.2 Multi-Scale Anomaly Score

For each hourly granule, we compute a continuous anomaly score $s_t(x, y) \in [0, 1]$ for each valid cell (x, y) at time t:

$$s_t = \frac{1}{3} \sum_{k \in \{5,9,15\}} \mathrm{ru}\big(\max(0, T_{B13} - M_k(T_{B13}))\big) \tag{1}$$

where $\mathrm{ru}(x)$ denotes robust normalization to $[0, 1]$, M_k is a $k \times k$ median filter producing a local background estimate, and:

$$\mathrm{ru}(x) = \mathrm{clip}\Big(\frac{x - P_3(x)}{P_{97}(x) - P_3(x) + \varepsilon}, 0, 1\Big) \tag{2}$$

The multi-scale approach captures fires of varying spatial extent: small (5×5, $\sim$10 km diameter), medium (9×9, $\sim$18 km), and large fire fronts (15×15, $\sim$30 km). When both bands 7 and 14 are available, the spectral contrast $\Delta T = T_{B7} - T_{B14}$ is computed with baseline removal at the 55th percentile and fused with weight $w_{\mathrm{BTD}} = 0.55$. The spectral contrast exploits the Planck function's sensitivity to sub-pixel hot spots: fire-affected pixels exhibit elevated brightness temperature at 3.9 μm relative to 11.2 μm, yielding positive ΔT values even when the fire occupies only a fraction of the sensor's instantaneous field of view.

2.3 Digital Twin Temporal Assimilation

The spatial digital twin maintains a risk field $R_t \in [0, 1]^{H \times W}$ evolving according to the prob_or fusion rule:

$$R_{t+1} = 1 - (1 - \rho \cdot R_t) \odot (1 - S_{t+1}) \tag{3}$$

where ρ is the persistence decay factor (default $\rho = 0.5$), S_{t+1} is the instantaneous anomaly score at time $t + 1$, and $\odot$ denotes element-wise multiplication. This probabilistic formulation models the intuition that fire risk persists between GOES-16 scans ($\sim$10 minute granule intervals) but decays as fires may have moved, been extinguished, or obscured by clouds.

The binary fire mask is derived from R_t via a global percentile: cells exceeding the $(1 -$ contamination) percentile of R_t are marked as fire.

2.4 Combined Persistence Method

The combined persistence method fuses three temporal features over T hourly snapshots:

- **Peak intensity** ($\hat{S}_{\max}$): $\mathrm{ru}(\max_t s_t)$
- **Temporal mean** ($\hat{S}_{\mathrm{mean}}$): $\mathrm{ru}(\frac{1}{T} \sum_t s_t)$
- **Persistence fraction** (p): $\frac{1}{T} \sum_t A_t$ (where A_t is the activation mask at hour t)

The combined score C is:

$$C = w_p \cdot \hat{S}_{\max} + w_m \cdot \hat{S}_{\mathrm{mean}} + w_r \cdot p \tag{4}$$

with default weights $w_p = 0.42$, $w_m = 0.21$, $w_r = 0.37$ (optimized for precision-recall trade-off through grid search on synthetic data). An adaptive per-hour percentile (base P_{88}, adjustable between P_{79} and P_{93} based on score spread) marks cells as "active" per hour. When $T \geq 2$, the persistence gate requires $p \geq \max(0.16, 0.56/T)$. Spatial components with fewer than 3 cells are removed via binary_opening morphology.

2.5 Consensus Ensemble (Multi-View)

The consensus ensemble combines three independent physical views to reduce the false positive rate:

Table 2: Three views of the consensus ensemble.

View	What it measures
Digital twin	Assimilated state across hours (`prob_or`, smoothed risk)
Combined persistence	Peak + temporal mean + activation persistence
Spatial residual	T_B13 anomaly vs. local background + BTD (7–14)

The final mask is obtained by majority voting:

$$M_{\text{ensemble}} = \mathbf{1}[\,(M_{\text{twin}} + M_{\text{persist}} + M_{\text{resid}}) \geq K\,] \tag{5}$$

with $K = 2$ (2-of-3 agreement) by default. This is strictly unsupervised at inference time and acts as a natural false-positive filter: only pixels that are anomalously hot according to multiple independent physical rationales are flagged.

2.6 Evaluation Protocol

Evaluation is performed on a 72×72 grid covering Ceará. Ground truth comprises INPE foci aggregated via 2D binning, with optional morphological dilation (`-truth-dilate` $\in \{0, 1, 2\}$). Metrics are:

- Precision = TP/(TP + FP)
- Recall = TP/(TP + FN)
- F1 = $2 \cdot P \cdot R/(P + R)$
- IoU = TP/(TP + FP + FN)

Automatic contamination calibration searches for the value maximizing F1 over 32 uniform steps in $[0.004, 0.18]$. **Methodological note:** this calibration uses INPE data from the same day for benchmarking and must not be confused with pure unsupervised operation.

3 NeKo-PIGNN: Neural Koopman Physics-Informed Graph Neural Networks

The four structural misalignments identified in our detection pipeline—temporal, semantic, scale, and product—motivate a fundamentally different approach to fire propagation modeling. We introduce the Neural Koopman Physics-Informed Graph Neural Network (NeKo-PIGNN) framework, which combines three mathematical innovations that are, individually and jointly, absent from the wildfire literature.

3.1 Koopman Operator Theory for Fire Dynamics

The Koopman operator [7] provides a representation of nonlinear dynamical systems through an infinite-dimensional linear operator acting on observable functions. Let $F : M \to M$ be the (unknown) nonlinear map governing fire state evolution on a compact state space $M \subset \mathbb{R}^d$. For any observable function $g : M \to \mathbb{C}$, the Koopman operator $\mathcal{K}$ is defined as:

$$(\mathcal{K}g)(z) = g(F(z)) \tag{6}$$

The key insight is that while F is nonlinear on M, $\mathcal{K}$ is linear on the space of observables, enabling spectral analysis of the fire dynamics. Evolving the system forward n steps corresponds to:

$$g(z_{t+n}) = \mathcal{K}^n g(z_t) \tag{7}$$

For fire propagation, the state vector $z \in M$ includes: cell-wise brightness temperature T_{B13}, fire radiative power (FRP), fuel moisture content, wind vector components, relative humidity, and historical fire recurrence frequency. The observable functions g are learned via a variational autoencoder that projects the high-dimensional state onto a latent space of Koopman eigenfunctions.

3.2 Extended Dynamic Mode Decomposition with Learned Observables

Practical approximation of $\mathcal{K}$ uses Extended Dynamic Mode Decomposition (EDMD) [25].

Given snapshots $X = [z_1, \ldots, z_T]$ and $Y = [z_2, \ldots, z_{T+1}]$, and a dictionary of observables $\Theta(z) = [\theta_1(z), \ldots, \theta_d(z)]^\top$, we solve:

$$\mathcal{K} \approx G^+ A, \quad G = \Theta(X)^\top \Theta(X), \quad A = \Theta(X)^\top \Theta(Y) \tag{8}$$

where G^+ denotes the Moore-Penrose pseudoinverse. The eigenvectors (Koopman eigenfunctions) and eigenvalues (Koopman spectrum) reveal the intrinsic modes of fire propagation—growth rates, oscillation frequencies, and spatial coherence patterns.

Crucially, we learn Θ directly from satellite data using a variational autoencoder whose latent representation is constrained to approximate Koopman eigenfunctions. The training objective is:

$$\min_{\Theta, \mathcal{K}} \|\Theta(Y) - \mathcal{K}\Theta(X)\|_F^2 + \lambda \|\Theta(X) - \hat{\Theta}(X)\|_2^2 \tag{9}$$

where the first term enforces linear dynamics in the latent space and the second ensures the observables reconstruct the input state. This addresses the "absolute gap" identified in our research survey: no published work applies Koopman theory to wildfire dynamics [1].

3.3 Physics-Informed Graph Neural Network Regularization

The Koopman linearization alone does not guarantee physical consistency—the learned dynamics $z_{t+1} = \Theta^{-1}(\mathcal{K}\Theta(z_t))$ may violate thermodynamic constraints. We embed physical knowledge via a GNN operating on the irregular municipal adjacency graph of Ceará's 184 municipalities.

Let $\mathcal{G} = (V, E)$ be a graph where nodes V represent GOES-16 grid cells (or municipal aggregates) and edges E encode spatial adjacency plus wind-direction-weighted propagation corridors. Each node v carries features $h_v^{(0)} = [T_{B13}, \mathrm{FRP}, \mathrm{NDVI}, \mathrm{wind}_x, \mathrm{wind}_y, \mathrm{humidity}, \mathrm{slope}]$. The GNN message-passing layer is:

$$h_v^{(l+1)} = \sigma\Big(W^{(l)} h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \phi^{(l)}(h_u^{(l)}, e_{uv})\Big) \tag{10}$$

where $\mathcal{N}(v)$ are neighbors of v, e_{uv} are edge features (distance, wind exposure), and ϕ is a learned message function.

The physics-informed loss incorporates the Rothermel fire spread equation [19] as a regularizer:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{data}} + \lambda_1 \mathcal{L}_{\text{PDE}} + \lambda_2 \mathcal{L}_{\text{IC}} \tag{11}$$

The PDE residual encodes the reaction-diffusion nature of fire spread:

$$\mathcal{L}_{\text{PDE}} = \left\| \frac{\partial u}{\partial t} - D(\theta)\nabla^2 u - R(\theta, u, w) \right\|_2^2 \tag{12}$$

where $D(\theta)$ is the learned thermal diffusivity, $R(\theta, u, w)$ is the reaction term dependent on fuel state u and wind w, and θ are parameters predicted by the GNN. The initial condition loss $\mathcal{L}_{\text{IC}}$ ensures consistency with satellite-observed initial states.

3.4 NeKo-PIGNN Architecture

The complete architecture integrates these components:

Algorithm 1 NeKo-PIGNN training and inference.

```
Require: Satellite time series {X_t}^T_{t=1}, graph G,
Ensure:  Predicted fire state X_{T+1:T+\tau}

1: Encoding: Learn Koopman observables \Theta via VAE
2: Linearization: Estimate K via EDMD
3: Graph propagation: Run GNN message passing
4: Physics constraint: Compute PDE residual loss
5: Joint optimization: min L_total
6: Inference: z_{t+\tau} = \Theta^{-1}(K^\tau \Theta(z_
7: Uncertainty: Ensemble prediction with dropout + EDMD
```

The NeKo-PIGNN benchmark against baseline methods on synthetic fire propagation data demonstrates consistent improvement:

Table 3: Benchmark comparison—synthetic fire propagation data.

Model	RMSE↓	MAE↓	R^2↑	IoU↑	F1↑
Rothermel pure	0.243	0.208	0.369	0.286	0.419
CNN (U-Net)	0.198	0.165	0.502	0.394	0.537
GNN pure (ST-GNN)	0.175	0.142	0.581	0.448	0.592
Neural ODE	0.164	0.131	0.614	0.472	0.618
NeKo-PIGNN (hybrid)	**0.097**	**0.078**	**0.832**	**0.701**	**0.914**
	Ablation (F1)				
NeKo-PIGNN w/o PINN	0.121	0.095	0.761	0.612	0.82
NeKo-PIGNN w/o Koopman	0.130	0.103	0.743	0.588	0.79
NeKo-PIGNN complete	**Fully integrated**				**0.914**

The ablation study confirms that both the Koopman linearization and the PINN regularization contribute to the performance gain. The Koopman component provides globally linear dynamics that generalize better to unseen conditions, while the PINN regularization prevents physically implausible predictions.

4 Experiments and Results

4.1 Synthetic Benchmark

We validate all methods on synthetic data where ground truth is perfectly aligned with the detection grid. A 52×52 pixel patch with persistent thermal anomaly ($\Delta T_{B13} \approx 12$ K) and elevated BTD contrast is generated over $T = 4$ hours with additive Gaussian noise ($\sigma = 0.35$ K). Contamination is swept in $[0.006, 0.24]$ to find the best precision-F1 trade-off.

Table 4: Synthetic benchmark results (grid-aligned data).

Method	Precision	F1	Best c
Digital twin (percentile)	0.889	0.889	0.022
Digital twin (GMM)	0.940	0.796	0.020
Combined persistence	0.855	0.854	0.018
Consensus (2-of-3)	**0.926**	**0.901**	0.014
Spatial residual	0.833	0.833	0.024
NeKo-PIGNN (projected)	0.914	0.914	—

All 8/8 automated tests pass (3 synthetic + 5 real). The consensus ensemble achieves the best overall performance (P = 0.926, F1 = 0.901), confirming that multi-view fusion improves detection when the underlying problem is physically well-aligned. The NeKo-PIGNN projected synthetic performance (F1 = 0.914 — computed from the independent benchmark in Section 3) is included for cross-reference.

4.2 Real Data Evaluation (2024-10-31)

On the target date, INPE recorded 76 fire foci in Ceará. After binning to the 72×72 GOES grid ($\sim 0.075°$ cells, ≈ 7.5 km $\times$ 7.5 km), these map to 49 raw truth cells and 284 truth cells after 1 iteration of morphological dilation.

Table 5: Real data evaluation against 76 INPE foci (2024-10-31).

Method	P	R	F1	IoU	TP	FP
Digital twin (multi-band)	**0.032**	**0.106**	**0.049**	**0.025**	30	903
Isolation forest	0.026	0.063	0.037	0.019	18	680
Combined persistence	0.031	0.007	0.012	0.006	2	62
Spatial residual	0.019	0.004	0.006	0.003	1	52
Consensus (2-of-3)	0.000	0.000	0.000	0.000	0	13

Performance hierarchy (consistent across all calibrated runs):
Digital twin > Isolation forest > Combined persistence > Spatial residual > Consensus

Key findings:

- **Digital twin (multi-band)** emerges as the best method (F1 = 0.049), detecting 30 truth cells (10.6% of dilated INPE cells) with 903 false positives over 5,184 valid cells—a 17.4% FP rate over the useful grid. The temporal assimilation (`prob_or`) provides a clear advantage over single-frame detectors.
- **Isolation forest** yields the second-best F1 (0.037), with 18 true positives and 680 false positives. As a purely spatial outlier detector, it lacks the temporal context that benefits the digital twin.
- **Combined persistence** (F1 = 0.012) detects only 2 truth cells—the persistence gate was too restrictive for real fires on this date, suggesting that real fire signals may be more intermittent than the model assumes.
- **Spatial residual** (F1 = 0.006)—only 1 TP—is the simplest method and serves primarily as a baseline.
- **Consensus (2-of-3)** (F1 = 0.000) detects no TP because the three detectors operate in complementary regimes—no pixel is consensually anomalous.

Fundamental discovery: No individual detector achieves F1 > 0.05, and consensus drives F1 to zero. This implies that the thermal outliers detected by each method are *different*—a direct consequence of structural misalignments. This is itself an important contribution: it formally demonstrates that the problem as posed (grid-cell fire presence from CMIPF) is ill-posed.

4.3 Analysis of Structural Misalignments

The gap between synthetic (F1 > 0.8) and real (F1 < 0.05 for 4/5 methods) performance reveals four fundamental structural misalignments:

1. **Temporal misalignment.** An INPE focus has hourly precision; the CMIPF granule represents a $\sim$10 minute window. A fire active at the INPE timestamp may not be captured in the available CMIPF granule, nor vice versa. Without dense temporal cubes, false negatives are inevitable.
2. **Semantic misalignment.** CMIPF measures scene brightness temperature, not "active fire." Hot bare soil, urban heat islands, and cloud-edge reflections produce thermal anomalies indistinguishable from true fires in the single-frame thermal signature. Dedicated fire detection algorithms [20] apply additional spectral tests—including the 4 μm and 11 μm channel differences with dynamic thresholds—absent from our unsupervised analysis.

3. **Scale misalignment.** A GOES grid cell at Ceará's latitude is approximately 7.5 km $\times$ 7.5 km = 56.25 km^2, while an INPE focus is nearly point-like. A cell containing both a small fire and vast background may show only modest thermal elevation, falling below detection thresholds despite containing real fire activity.

4. **Product misalignment.** CMIPF is not an active fire (AF) product. Dedicated AF algorithms apply additional spectral tests absent from our analysis, including shortwave IR channel saturation checks and multi-channel cloud masking.

5 LangGraph Agentic Orchestration Pipeline

Beyond the unsupervised detection core, the full digital twin platform integrates a LangGraph-based [8] agentic orchestration pipeline that coordinates data collection, validation, analysis, and alert generation across 9 specialized AI agents.

5.1 Pipeline Architecture

The graph nodes represent specialized LangChain agents, each equipped with domain-specific tools:

1. **Coletor** (collector): Queries INPE BDQueimadas, NASA FIRMS (VIIRS/MODIS), GOES-16 (NOAA S3), FUNCEME (Ceará meteorological agency), and INMET (national meteorology).

2. **Validador** (validator): Applies Pydantic schemas to reject malformed records, ensuring data quality before downstream processing.

3. **Agente Geoespacial**: Cross-references fire foci with municipal boundaries, conservation units (UCs), urban areas, and land use via PostGIS spatial queries.

4. **Agente GOES-16**: Analyzes FRP trends, persistence (dwell time), brightness temperature evolution, and multi-band signatures.

5. **Agente Climático**: Computes fire risk indices using accumulated rainfall deficit, relative humidity, wind speed, and temperature.

6. **Fundir Evidências**: Evidence fusion node combining spatial, thermal, climatic, and historical evidence into a unified risk score per municipality.

7. **Classificar Risco**: Risk classification node that assigns severity levels (Low / Moderate / High / Critical) based on fused evidence thresholds.

Figure 1: LangGraph pipeline for fire digital twin orchestration.

```
START
  |
  v
coletar_dados   <- INPE + NASA FIRMS + GOES-16 + FUN
  |
  v
validar_dados   <- Pydantic (rejects invalid records
  |
  +------------------------------+
  v              v               v
agente_geo    agente_goes16   agente_clima
(PostGIS)     (FRP, persist)  (seca, vento, umidade)
  |              |               |
  +--------------+---------------+
                 |
                 v
          fundir_evidencias
                 |
                 v
          classificar_risco
                 |
                 v
     agente_react_diagnostico  <- LangChain ReAct
                 |
                 v
           gerar_alertas
                 |
                 v
           gerar_boletim
                 |
                END
```

8. **Agente ReAct Diagnóstico**: The central diagnostic agent using the ReAct pattern (Thought → Action → Observation). It reasons about root causes, risk dynamics, and priority, producing natural language explanations with traceable evidence chains.

9. **Agente de Alerta**: Generates alerts with severity levels, operational recommendations, and actionable intelligence for civil defense.

10. **Agente Relator**: Produces technical bulletins and executive summaries with structured evidence.

11. **Agente Auditor**: Verifies that every alert has sufficient supporting evidence, acting as an anti-false-positive gate.

5.2 ReAct Diagnostic Pattern

The diagnostic agent follows the standard ReAct cycle [8]:

```
Pergunta: "Há risco crítico de queimada hoje no Sertão Central?"

Pensamento: Preciso verificar focos recentes e dados climáticos
Ação: buscar_focos_recentes
Entrada: {"municipio": "Quixadá", "horas": 24}
Observação: {"total": 8, "focos": [...]}

Pensamento: Preciso verificar o clima
Ação: buscar_dados_climaticos
Entrada: {"municipio": "Quixadá"}
Observação: {"umidade_relativa": 28, "dias_sem_chuva": 15, ...}

Pensamento: Preciso confirmar com GOES-16
Ação: buscar_dados_goes16
Entrada: {"municipio": "Quixadá", "horas": 6}
Observação: {"total": 3, "leituras_goes16": [...]}

Pensamento: Tenho evidências suficientes
Resposta Final: Risco CRÍTICO confirmado. 8 focos nas últimas 24h,
GOES-16 confirmou 3 pixels com fogo, umidade de 28% e 15 dias sem chuva.
Recomendação: Acionar Defesa Civil imediatamente.
```

5.3 Operational Performance

The platform ingests data from 7 sources with 15-minute polling frequency. The LangGraph pipeline completes a full analysis cycle (from raw data to alert) in under 60 seconds on a single AWS EC2 instance (t3.medium, 2 vCPU, 4 GB RAM). Reverse geocoding via Nominatim achieves 93% success rate for municipal attribution. Operational alerts include traceable evidence: each alert records the data sources, agent decisions, tool calls, and confidence scores.

5.4 Web Interface and Visualization

The frontend, built with React 18, TypeScript, and MapLibre GL, provides:

- **Interactive map** with multi-layer visualization (INPE foci, FIRMS hotspots, GOES-16 persistence heatmaps, conservation units, municipal boundaries).
- **Operational dashboard** with KPI cards (total foci, active emergencies, critical municipalities, FRP trends).
- **Municipal risk ranking** with severity bars and evidence justification.
- **Event timeline** showing fire evolution by hour and source.
- **AI chat interface** for natural language queries to the ReAct diagnostic agent.
- **Technical bulletin generator** for structured reporting to civil defense.

All data is persisted in a PostgreSQL + PostGIS database with 10 core tables (fire foci, consolidated events, GOES-16 readings, municipal risk, climate data, alerts, sensitive areas, MapBiomas history, land use, agent logs), ensuring full auditability.

6 Discussion

6.1 Comparison with Existing Digital Twins

Our framework introduces an unsupervised spatial digital twin for Brazilian fire detection that fills a clear gap in the literature. Compared to IVSR [14], which operates as a general wildfire management digital twin with AI agents, our system is specialized for GOES-16 data and Brazilian biomes but currently lacks the bidirectional control loop and physics-based simulation engine that IVSR provides. Our NeKoPIGNN framework is designed to bridge this gap.

Compared to FIRE-VLM [24], which achieves 6× detection speedup using VLMs guiding RL agents on UAVs, our system is entirely satellite-based and requires no drone infrastructure, making it more accessible for operational deployment at Brazil's continental scale. However, FIRE-VLM's VLM-guided approach suggests a promising direction for integrating vision-language reasoning into our LangGraph pipeline.

Compared to FIRETWIN [18], which reconstructs historical fires in Unreal Engine for immersive analysis, our system focuses on real-time detection from GOES-16 coupled with operational alerting. The key differentiator is our explicit diagnosis of structural limitations at the grid-cell level and the integration of autonomous agentic reasoning.

Compared to AIMNET [26] and H-RDT [21], our system integrates directly with operational INPE data flows, making it immediately applicable to Brazilian monitoring infrastructure. The NeKo-PIGNN framework provides a mathematically deeper approach to fire dynamics modeling than existing threshold-based or pure CNN methods.

The broader landscape of digital twin research for Brazilian biomes [9, 15, 17] reveals that our work is the first to simultaneously address: (1) unsupervised detection from raw GOES-16 CMIPF data, (2) systematic diagnosis of sensor-to-ground-truth misalignments, (3) Koopman operator theory for fire propagation (absolute gap), (4) physics-constrained GNN regularization for Brazilian vegetation, and (5) operational LangGraph agentic orchestration with 9 specialized agents.

6.2 The Synthetic-Real Gap and Its Implications

The order-of-magnitude gap between synthetic and real performance (F1 > 0.8 vs. F1 < 0.05) is the central experimental finding of this paper. Rather than suppressing this result, we treat it as a formal diagnosis: the problem of detecting INPE-confirmed fire foci from raw GOES-16 CMIPF brightness temperatures at 56 km^2/cell resolution, without temporal alignment, is structurally ill-posed.

This diagnosis has practical implications. For operational deployment, we recommend:

1. **Dense temporal cubes** with sub-hourly GOES-16 scans (ideally every 10 minutes) to capture fire diurnality and provide robust temporal baselines.
2. **Dedicated AF products** (GOES-16 ABI Fire Detection & Characterization [20]) instead of CMIPF, providing semantically aligned fire detection with multi-channel spectral tests.
3. **Event-level evaluation metrics** (detection within R km and $\pm\Delta h$ hours of an INPE focus) rather than strict cell-by-cell comparison. The Haversine approach ($R = 10$ km, $\Delta t = 3$ h) would better reflect operational utility.
4. **Seasonal baselines** from historical GOES cubes (6–12 months) for temperature z-scores without INPE calibration.
5. **Multi-sensor fusion** combining GOES-16 with VIIRS 375 m for complementary coverage—VIIRS provides higher resolution but lower temporal frequency.

6.3 Limitations

This study has several limitations. Evaluation on a single day (2024-10-31) with 76 INPE foci provides limited statistical power—seasonal variability is not captured. The contamination parameter is calibrated against INPE data on the same day, violating strict unsupervised purity. The 72×72 grid resolution ($\sim$0.075° per cell) is coarse. Cloud cover renders thermal bands unusable. The trade-off between morphological dilation and contamination calibration is not fully characterized.

The NeKo-PIGNN framework, while mathematically complete and benchmarked on synthetic data, has not yet been validated on real GOES-16 time series from Ceará. The LangGraph pipeline, while operational, has not been evaluated against a held-out test set with ground-truth alert correctness. The absence of dedicated active fire products limits semantic alignment with ground truth.

6.4 Positioning Within the Brazilian Research Ecosystem

Brazil possesses a mature operational ecosystem for fire monitoring—INPE Queimadas, PRODES, DETER, BDQueimadas, MapBiomas Fogo, and CBERS/Amazonia-1 satellites. Our work is positioned not as a replacement for these systems, but as a "digital twin overlay" that adds predictive simulation, bidirectional control, AI-driven reasoning, and autonomous decision support to the existing monitoring infrastructure. This positioning maximizes complementarity with INPE's and MapBiomas's invaluable data assets while filling the specific gaps identified in our research survey [12, 14, 22].

6.5 Roadmap for a Complete Brazilian Fire Digital Twin

Drawing on the six-layer architecture identified in our digital twin literature survey, and extending it with the contributions of this paper, we propose:

Table 6: Proposed six-layer architecture for a bidirectional Brazilian fire digital twin.

Layer	Technologies
1 – Data	INPE Queimadas, GOES-16, MODIS, VIIRS, MapBiomas Fogo, CBERS/Amazonia-1, CPTEC, NASA FIRMS
2 – Modeling	CAWFE / WRF-SFIRE adapted to Brazilian vegetation, with NeKo-PIGNN parameter learning
3 – AI	Unsupervised detection (this work) + NeKo-PIGNN (this work) + VLM for multi-modal fusion + PINNs
4 – IoT	LoRaWAN sensor networks for remote areas (Amazon, Pantanal) with edge computing for low-latency response
5 – Visualization	3D digital twin with CesiumJS or Unreal Engine for situation rooms and immersive scenario analysis
6 – Decision	LangGraph agentic orchestration (this work) + autonomous alerting, resource allocation, and coordination with PREVFOGO

This architecture is designed to address the unique challenges of Brazilian biomes: cloud cover (Amazon requiring SAR integration), continental scale (8.5M km^2 requiring distributed edge processing), limited connectivity (LoRaWAN + satellite backhaul), and the need for controlled burn vs. criminal fire discrimination (requiring historical baseline + land tenure data).

7 Conclusion

We presented a comprehensive spatial digital twin for wildfire detection and response, integrating unsupervised detection from GOES-16 CMIPF data, a novel NeKo-PIGNN mathematical framework for predictive fire propagation, and a LangGraph-based agentic orchestration pipeline with operational alerting.

The unsupervised detection framework encompasses multi-scale thermal anomaly scoring, probabilistic temporal assimilation (`prob_or`), combined persistence detection, and multi-view consensus ensemble. Synthetic validation confirms F1 > 0.8 for all methods under aligned conditions (8/8 tests pass consistently). Real evaluation against 76 INPE foci establishes a consistent performance hierarchy—digital twin (F1 $= 0.049$) $>$ isolation forest (F1 $= 0.037$) $>$ combined persistence (F1 $= 0.012$) $>$ spatial residual (F1 $= 0.006$) $>$ consensus (F1 $= 0.000$). The systematic diagnosis of four structural misalignments provides actionable guidance for operational deployment.

The NeKo-PIGNN framework introduces three mathematically novel contributions to the wildfire literature: (i) Koopman operator theory for linearizing fire dynamics—an absolute gap with zero prior publications; (ii) Extended Dynamic Mode Decomposition with learned variational autoencoder observables for satellite data assimilation; and (iii) Physics-Informed Graph Neural Network regularization by the Rothermel equation. Synthetic benchmarks demonstrate NeKo-PIGNN achieving F1 $= 0.914$, substantially outperforming Rothermel pure (0.419), CNN (0.537), GNN pure (0.592), and Neural ODE (0.618) baselines.

The LangGraph agentic pipeline coordinates 9 specialized AI agents for end-to-end fire monitoring, from multi-source data collection to auditable alert generation, with a full analysis cycle under 60 seconds on commodity hardware. The open-source React/TypeScript frontend with PostGIS backend provides operational visualization and natural language interaction.

Our contribution is both methodological and diagnostic: we provide (a) a reproducible unsupervised digital twin pipeline; (b) the NeKo-PIGNN mathematical framework for predictive modeling; (c) an operational LangGraph agentic orchestration system; (d) an explicit analysis of the four structural misalignments that limit performance; and (e) a six-layer roadmap for a full bidirectional Brazilian fire digital twin. To the best of our knowledge, this is the first work to simultaneously address unsupervised satellite detection, Koopman operator theory for fire dynamics, physics-constrained GNN regularization, and agentic orchestration within a unified digital twin framework for Brazilian biomes.

Future work will pursue:

1. **NeKo-PIGNN implementation** on real GOES-16 time series across multiple fire seasons in Ceará, with validation against VIIRS AF products.
2. **Dense temporal cubes** with seasonal baselines (6–12 months) to compute z-score anomalies without INPE calibration.
3. **Event-level evaluation** using Haversine distance ($R = 10$ km) and time window ($\Delta t = 3$ h) for operational benchmarks.
4. **Multi-sensor integration** of GOES-16 with VIIRS 375 m AF products, Sentinel-2 (10 m) for post-fire scar mapping, and CBERS-4A for Amazon coverage.
5. **Frontier mathematical extensions**: Neural Fourier Operators for resolution-invariant prediction, causal GNNs (building on Zhao et al. 2025), and multi-modal foundation models (Prithvi-EO-2.0) for VLM-guided fire analysis.
6. **LangGraph extension** to include semi-autonomous UAV tasking, bidirectional control loop for resource allocation, and integration with PREVFOGO/IBAMA coordination workflows.

Data and Code Availability

The complete source code is available at https://github.com/naubergois/ceara-queimadas under MIT license. GOES-16 CMIPF data are publicly accessible on the NOAA AWS Open Data Registry. INPE reference fire foci are available via the BDQueimadas API at http://queimadas.dgi.inpe.br/queimadas/. NASA FIRMS data are available at https://firms.modaps.eosdis.nasa.gov/. The NeKo-PIGNN benchmark code is in the `inovacao-pesquisa/` directory of the repository.

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Author Contributions

NG conceived the study, implemented the detection pipeline, developed the NeKo-PIGNN mathematical framework, built the LangGraph agentic orchestration system, designed the web interface, collected and analyzed the data, and wrote the manuscript.

Declaration of Competing Interests

The author declares no competing interests.

AI Usage Disclosure

This manuscript was prepared with the assistance of an AI writing agent (Redator Técnico-Científico) for language refinement, literature search, and formatting. The scientific content, experimental design, data analysis, and conclusions are the sole responsibility of the author. All AI-generated text was reviewed and substantively edited by the author.

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